# How a reported number is produced

From the frozen specification to the macro the prose cites, with no hand-copied value anywhere on the path

**Frozen specification**

spec/SPECIFICATION.md fixes the API, state machine, data structures, and concurrency model, before any code is written.

**Four build variants**

Two languages crossed with two allocators. Matched optimisation levels; both sides pin the same shard and intra-shard hashes.

**Deterministic driver**

A fixed SplitMix64 draw order and a per-slot expected-state model, so both versions walk identical code paths.

**Service and capture**

Eight shards, one owning thread each, 1024-slot queues. HDR histogram timed from the intended send time, not the actual.

**Repeated 20 times, with the four variants interleaved inside each repeat**

14 offered-load levels up to 784 kops/s give 1120 measured runs; interleaving makes the interference common to a block, so it cancels.

**Raw measurement logs**

bench/results/*.jsonl, with the environment record from capture_env.sh kept alongside each result set.

**bench/plot_figures.py**

Median with a bootstrap 95% confidence interval, an exact paired permutation test, and the Holm correction.

**Generated artifacts**

figures/*.pdf
figures/tab_*.tex
figures/gen_macros.tex

**Thesis LaTeX source**

The prose refers to the generated macros by name, so text and data cannot drift.

**Nothing in the evaluation is hand-copied**

bench/refresh_thesis.sh reruns the analysis and rebuilds the PDF, so a fresh campaign updates every figure, table, and quoted number in one step.

A third party can walk this path without proprietary code: the specification and the harness are themselves published artifacts.